



Knowing what is actually true has always been a hard problem but in the age of the internet and LLMs, it is almost impossible to know what is true. One possible solution is digital signatures and an advanced reputation system. Such a system would need to be developed. An alternative, and this is my focus now, is to generalise prediction markets to include bets not mainly on future uncertain events but on claimed facts. Current prediction markets allow people to bet on a relatively small number (maybe thousands or tens of thousands) of statements. I want to be able to make bets on potentially millions (or more) of statements, very ordinary ones, such as what the opening time of a particular supermarket is, or statements found in general common-sense databases, such as wikidata, openstreetmap, and others. There are several potential problems with this approach, and one of these is that to make a very specific bet, you need to find a counterparty to bet with, and you may not find such a counterparty. One potential solution would be to make the bets very efficient, so that making such bets would become easy in large quantities. Another would be to batch many statements together (that all of them are true). Yet another would be to not just make individual bets but a bets embedded in a knowledge graph of other bets, which may give liquidity by generalisation (betting against some combination of implications of each bet). These are my starting ideas, and my name is Peter Foldiak.

I also started discussions about this topic with my friend Daniel A. Nagy, who pointed out further problems with bets. In theory, the price of bets on prediction markets would be equal to the probability of events. But he observed that this is often not the case, especially for events with probabilities close to 1, and bets on longer time scales. He says this is caused by such bets not being capital efficient, i.e. if you have an even with 99% probability, you would have to pay 99 cents in the hope of winning 1 cent a year from now. Such a bet is not worth making, as you could earn more yield on your 99 cents. So he suggested that instead of buying a "yes" bet for 99 cents, we should be able to somehow short a "no" bet, for 1 cent instead. But as far as we know such shorting bets don't really exist. We would need a mechanism (eventually supported by smart contracts) to be able to make leveraged bets, which would allow us to make those much more capital efficient shorts on the opposite position.

We don't really know how to do that but would be interested in ideas on this. I also suggested that it would be great to be able to pool our stakes in many simultaneous bets. So if I have could do some kind of "fractional reserve" pooling, then a smaller capital could be used to make a large number of bets. So when some of the bets are lost, the reserve would be used, and we assume that not all bets would fail at exactly the same time. Then at ETHPrague 2026 we attended the interesting prediction markets workshop of Krzysztof Paruch of Token Engineering Labs and Daniel and I suggested starting a project aimed at solving these problems as Krzysztof has expertise in prediction markets. We will be joined by Mark B Richardson, CEO, Head of research and protocol design at Bancor. We think the first problem to solve is the leverage and shorting problem for bets. I am attaching his proposal.

Please analyse the topics raised here and express an honest opinion. We would also appreciate any previous work you can find and your own ideas or suggestions for how to get started and what low hanging fruits or promising directions you see.

The project, as you can see, is extremely ambitious and if we succeed would have huge implications for knowing what is true. It would also be an incentive for people to find false or inaccurate statement or data in knowledge bases and would even be a kind of funding mechanism for research on unpopular opinions. People could put their money where their mouths are. Please do a thorough analysis, listing sources.

Show less

Synthesized prediction market literature, identified core mechanisms, and mapped implementatio...

This is a genuinely interesting project, and I'll give you my honest assessment: the vision is sound, several pieces of it already exist in forms you may not have fully connected to your framing, and the hardest problems are not the ones the proposal currently emphasizes most. Let me go through it.

The core reframe I'd suggest: for millions of mundane facts, the right primitive is probably not a market — it's a bonded assertion

A continuous two-sided market per statement cannot work at the scale you describe. Millions of markets on supermarket opening hours will have zero traders, and even a subsidized automated market maker (Hanson's LMSR) has a bounded loss *per market* — the subsidy is the price you pay for information, and paying it millions of times over facts nobody disputes is exactly the capital inefficiency you're trying to escape.

But notice what a bet on a near-certain fact actually is: someone stakes capital on "this is true," and the market only becomes interesting if someone disagrees. That structure already exists and is battle-tested at scale — it's the **optimistic oracle**. In UMA's design, data is assumed true unless disputed; anasserter posts a bonded claim, a challenge window opens, and disputed claims escalate to a decentralized vote. Empirically, around 98.6% of requests settle without any dispute, and UMA's oracle has processed

roughly 60,000 assertions securing tens of billions in volume. **Token-curated registries** (Mike Goldin's TCR pattern, Kleros Curate) are the same idea applied to lists of facts: anyone can propose an entry by staking, entries face a challenge period, and challenges go to adjudication. Kleros's Generalized TCR explicitly lets anyone deploy a curated list of "just about anything" with financial incentives and dispute resolution.

UMA Blog + 4

So a "Wikidata verification market" is, in its base case, a TCR/optimistic-assertion system, and the reason this matters for your problem is profound: **the optimistic pattern is the capital-efficiency solution for the close-to-1 regime**. A dormant bonded claim is economically equivalent to a market whose implicit price is ~ 1 and which only "activates" (prices, trades, adjudicates) when challenged. Nobody locks 99 cents against every true fact; capital is only deployed at the point of disagreement, which is exactly where the information value is. I'd frame your project as: *take the optimistic-assertion primitive and give it the missing market properties* — continuous probabilistic prices where wanted, tradability, pooled underwriting, and graph-structured netting — rather than take prediction markets and try to scale them down to trivial facts.

Daniel's observation is correct and now has direct academic support

The 99-cent problem is real and recently formalized. A 2026 paper, "*When Certainty Is Not Worth It: Capital Lock-Up and Settlement Discounting in Prediction Markets*" (arXiv:2605.31431), argues exactly Daniel's point: the price-as-probability mapping is incomplete for collateralized, oracle-settled markets, and a substantial share of apparent long-horizon miscalibration — 48-88% of the near-certainty horizon gradient in their estimates — reflects mechanical settlement discounting from locked collateral rather than informational error. There's also a large empirical literature on the favorite-longshot bias showing high-probability outcomes are systematically underpriced (Snowberg & Wolfers 2010; Ottaviani & Sørensen 2009), and recent analysis of the full Polymarket trade tape confirms high-probability tokens exhibit positive realized returns, i.e. systematic underpricing. [arxiv](#) [arxiv](#)

The same paper points to the two known mitigations, both of which I'd call your genuine low-hanging fruit:

1. Yield-bearing collateral. If locked collateral earns the risk-free rate, the opportunity cost of holding a 99-cent position for a year largely vanishes. Yield-bearing collateral flattens the term structure of the settlement discount by reducing the opportunity cost of lock-up. This has been tried in practice (Gnosis's Omen/Presagio markets used sDAI), it requires no new mechanism-design theory, and for a verification market it means anasserter's stake on true facts is not dead capital — it's a yield-bearing deposit that additionally earns assertion fees. This alone changes the economics of staking on millions of true claims from "obviously irrational" to "an underwriting business." [arXiv](#)

2. Logical-structure-based collateral netting — which already exists in embryonic form as Polymarket's negRisk. This is the closest live implementation to your knowledge-graph idea, and I'd study it carefully. The NegRiskAdapter exploits the logical constraint that exactly one of a set of mutually exclusive outcomes resolves YES: a position of NO tokens across several outcomes is provably equivalent to YES tokens on the others plus cash, and the contract lets you convert between them. Polymarket reports up to 9.5x capital efficiency versus treating the same outcomes as independent markets. NegRisk handles only one logical relation (mutual exclusion within one event). Your knowledge-graph proposal is, in essence, *generalized negRisk*: if claim A implies claim B, then a stake on A should be able to serve as collateral for B; a set of claims connected by implications and exclusions defines a polytope of feasible outcomes, and collateral requirements should be computed against the worst case over that polytope rather than claim-by-claim. That is a crisp, publishable, implementable research question. [GitHub](#) [Polymarkets](#)

But here you hit a known wall: combinatorial markets are computationally hard

Your "liquidity by generalization" idea is Robin Hanson's combinatorial information market (Hanson 2003, 2007 — the founding papers, which introduce LMSR precisely to make betting on combinations of base events modular). The bad news is well established: Chen, Fortnow, Lambert, Pennock & Wortman showed LMSR pricing over combinatorial outcome spaces is #P-hard even for severely restricted betting languages. The good news is equally well established: tractability is recoverable by restricting structure. Bayesian-network and Markov-network factored representations admit efficient price updating, and have been used to run real combinatorial prediction markets (the SciCast/DAGGRE project ran a graphical-model market maker with thousands of linked questions), and Dudík, Lahaie and Pennock built a tractable combinatorial market maker using constraint generation that sits between treating securities as independent and full joint pricing. [ResearchGate + 2](#)

This maps beautifully onto your setting, because a knowledge graph is a factored structure. Wikidata-style claims have sparse, local logical dependencies (type constraints, functional properties, inverse relations), which is exactly the regime where graphical-model market makers work. A concrete research direction: define a restricted claim/implication language (say, Horn clauses over KB triples, or the constraint types Wikidata already encodes) for which both worst-case collateral computation and price propagation are polynomial. That would be a real contribution — the general problem is hopeless, but nobody has worked out the tractable fragment for knowledge-base verification specifically.

On leverage and shorting: mostly the wrong tool for your actual problem

Honest opinion: I think the proposal (and your framing) slightly over-indexes on "leverage" as the concept. Leveraged prediction markets now exist commercially —

dYdX ran election perpetuals with up to 20x leverage and standard margin fractions and liquidation, D8X launched leveraged orderbook markets on Polymarket outcomes, and there are newer entrants like HyperOdd, plus protocols like Gondor for borrowing against Polymarket positions without closing them. But margin-and-liquidation leverage has a property that's actively bad for your use case: a leveraged position can be forcibly liquidated on the price path before resolution. For a near-certain, long-dated fact claim, path-dependent liquidation means a manipulator who can temporarily move a thin market can liquidate honest verifiers — this degrades rather than improves information quality, and thin markets on obscure facts are maximally manipulable. Also note it doesn't remove the collateral, it just moves it to a lender who charges for it; you've converted an opportunity cost into an explicit funding rate.

Dydx + 3

What you actually want for "shorting the NO at 1 cent" is not perp-style leverage but **underwriting**: someone guarantees the near-certain outcome against a pooled reserve, collects the small premium, and is exposed to the rare loss. That's insurance, and it's your fractional-reserve idea — which brings me to:

The pooled-reserve idea is sound, and the right literature is actuarial, not DeFi

One reserve backing many partially collateralized correlated exposures is portfolio margining (SPAN, in TradFi) and insurance reserve sizing (Cramér-Lundberg ruin theory, solvency-capital frameworks). Existing crypto analogues worth studying: Kleros jurors stake once and that stake secures many simultaneous cases; UMA's entire oracle secures billions with modest bonds because the bond prices the cost of forcing a dispute rather than the value of the market, with the challenge window and escalation layer carrying the real economic weight; Nexus Mutual runs pooled underwriting for smart-contract risk. Your design question — how large a reserve for N claims with correlation structure ρ , dispute rate d , and adversarial claim selection — is a well-posed actuarial problem and a good WP4 deliverable. [Crypto News](#)

But the honest warning: **the failure correlations in a fact market are not like insurance correlations**. Three specific dangers:

1. **The adjudicator is a common risk factor.** Every claim in the pool settles through the same oracle/court. An oracle capture event (and UMA has had contested episodes — the crypto.news piece notes the known theoretical flaw that the profitable voting strategy is voting with the expected majority, not the truth, and a large holder can make the majority whatever they need) is a simultaneous shock to all positions. Diversification across claims does not diversify oracle risk. Your reserve model must price this as a systemic, not idiosyncratic, term. [Crypto News](#)
2. **Adverse selection is the default, not the exception.** In insurance, policyholders don't choose which houses burn. In a fact market, challengers *only* attack claims where the pool is wrong, and a fractional-reserve pool staking blindly across a

knowledge base is offering free options to anyone with better information. That's not a bug for the epistemics (it's exactly the error-finding incentive you want!) but it means the pool's economics must assume every challenge is informed. The pool is structurally the "dumb money" — it needs premium income (assertion fees, yield) calibrated to informed-challenger loss rates, not base-rate error rates.

3. **The lazy-verification / public-goods problem, which I consider the deepest issue in the whole project.** A bond only certifies truth if someone would profitably catch a lie. For a false statement about a small supermarket's hours, who spends the effort? The bounty must exceed the cost of verification research, per claim, forever. If it doesn't, "unchallenged" stops meaning "true" and the whole epistemic promise collapses silently — prices stay at ~1 not because the facts are verified but because verification isn't worth anyone's time. UMA works because its claims are few, high-value, and watched by thousands of position holders monitoring resolution; your setting is millions of low-value claims watched by nobody. Possible mitigations: verification subsidies funded by the knowledge base's consumers; random audit lotteries (amplify the expected bounty on a random subset); AI agents as cheap first-pass challengers — note UMA is already experimenting here, with an AI system that monitors and proposes market resolutions, currently at rough 78-90% accuracy against a human baseline of ~98% undisputed acceptance. I'd make "the economics of the marginal challenger" an explicit research question in WP4 — it's currently implicit at best. [Polysyncer](#) [UMA Blog](#)

Other prior work you should have on the reading list

Two strands the proposal doesn't mention and should. First, **information elicitation without verification / peer prediction** (Miller, Resnick & Zeckhauser 2005; Prelec's Bayesian Truth Serum; Kong & Schoenebeck's information-theoretic framework) — for facts where oracle settlement is too costly, you can pay agents based on agreement statistics rather than ground truth. Directly on point: Srinivasan, Karger & Chen's "Self-Resolving Prediction Markets for Unverifiable Outcomes" (EC 2025), which pays agents the negative cross-entropy between their prediction and a carefully chosen reference agent, and proves truthful reporting is a perfect Bayesian equilibrium. Be aware of the standard criticism that naive self-resolving designs are manipulable because there's no ground-truth reward offsetting a manipulator who trades the price wherever they want — the SKC mechanism addresses this in theory, but empirical robustness is open. A hybrid is attractive for your setting: self-resolution/peer consistency as the cheap default, with optimistic escalation to real adjudication as the backstop. [arXiv + 2](#)

Second, on market-making mechanics, Paradigm's recent design work: pm-AMM, an AMM derived specifically for outcome tokens' price dynamics, and distribution markets, for events whose outcome is a number rather than yes/no, letting traders express a full distribution. Given Mark Richardson is joining — Carbon's asymmetric-liquidity design experience is directly relevant to the question of what a market maker

for a claim that sits at 0.99 for years should look like (hint: almost all liquidity concentrated near 1, near-zero inventory risk until a dispute, which is very Carbon-like). Paradigm Paradigm

On Krzysztof's proposal itself

It's well-scoped and asks the right top-level questions; the capital-efficiency and shared-collateral emphasis in §3.3 is exactly right, and €10k for 2–4 weeks of two senior researchers is reasonable — honest, modestly priced discovery work with a clean payment structure. Three concrete suggestions:

1. **Name the literatures explicitly in WP2** so the review doesn't rediscover them: combinatorial market complexity (Chen et al. 2008; Hanson 2003/2007; Dudík-Lahaie-Pennock; the SciCast graphical-model market maker), IEWV/peer prediction/self-resolving markets (SKC 2023/2025), the capital-lockup pricing paper above, insurance ruin theory for reserve sizing, and the optimistic-oracle/TCR empirical record.
2. **Add the marginal-challenger economics** (my point 3 above) as a named research question. It's the question on which the whole "incentive to find false statements" vision hangs, and it's absent from Table 2.
3. **Pin the follow-up prototype to a machine-checkable claim domain.** Resolution ambiguity is the dominant practical failure mode of real prediction markets (Polymarket's disputes are almost all about wording, not facts). For a first prototype, choose claims with crisp, timestamped, formally specified semantics — e.g., Wikidata statements with external-ID references, or OSM objects with verifiable ground truth — rather than free-text facts. "The opening hours of shop X" is actually a hard claim: hours change, sources conflict, and the claim is reflexive (a bettor could get the sign changed). Versioned, timestamped claim semantics should be a first-class design element.

Where I'd start (my suggested sequencing)

If I were prioritizing: first, build the unifying theory note showing that optimistic assertions, TCRs, and prediction markets are points on one design spectrum (parameterized by when the market activates and who provides liquidity) — this reframing alone will organize everything else. Second, do the yield-bearing-collateral + assertion-fee economics for a static "stake on truth" pool, with ruin-theory reserve sizing including a systemic oracle-failure term — this is implementable with today's primitives and no new theory. Third, generalize negRisk to implication structures over a restricted claim language, starting with trees/DAGs where the worst-case collateral computation is tractable — this is the novel mechanism-design contribution. Fourth, and only then, decide whether continuous leveraged trading of claim tokens adds anything, with eyes open about path-dependent liquidation being an attack surface in thin markets. Simulate (cadCAD or similar) before writing Solidity, and treat the

marginal-challenger subsidy question as the go/no-go criterion for the whole vision: if honest verification of a mundane false claim can't be made profitable, the system will look healthy while quietly certifying nothing.

One last honest note on the grand vision. Even a perfect version of this system doesn't give you "what is true" — it gives you "what nobody found it profitable to dispute, under a specific adjudication procedure, at current bond and subsidy levels." That's genuinely valuable (it's more than you have now, for most databases), and the "next version

money where interesting. A comment on your section "The lazy-verification / public-goods problem": You say people may not challenge false information. You But prices are goods problem": You say people may not challenge false information. You demand, and may be right about not challenging random claims, but people would not social institu challenge random claims. Let's take the example of the supermarket epistemics. V opening hours. Who would want to know? It's the person who wants to shop at the supermarket, and he wants to make sure the supermarket is underneath i open when he gets there. Let's say he is willing to bet \$1 to a \$100 that it's Key sources: closed even though Openstreetmap tells him it is open. He would actually Chen, Fortnc prefer to lose this bet and uses it as information insurance. He gets to the Pennock on c shop and finds that it's closed. In this case he is highly motivated by the model marke \$100 to take a timestamped picture of the entrance in order to dispute Srinivasan, K the claim and get his \$100. So the false information is found by highly Snowberg & targeted people. Would this not work better than you said?

documentation; Goldin's TCR pattern and Kleros Curate; Polymarket's

NegRiskAdapter repo; Paradigm's pm-AMM and distribution markets posts.

Excavated insurance-as-verification mechanism, weighing elegance against practical complications

You're right, and this is a better design than the one I was implicitly critiquing. Let me say precisely what your reframing fixes, and then where the residual problems move to — because they don't disappear, they relocate.

What the insurance framing fixes. My lazy-verification objection assumed verification is a public good hunted by disinterested bounty hunters scanning millions of claims. Your version couples verification demand to *consumption*: the person who acts on the fact is exactly the person who (a) discovers the error at zero marginal cost — they're standing in front of the closed shop anyway, (b) holds a private financial claim on reporting it, and (c) is present at the right place and time to gather evidence. The public good (a corrected database entry for everyone) gets produced as a byproduct of a private transaction. That's a genuinely elegant mechanism — verification effort automatically allocates in proportion to how much each fact actually matters, which is arguably the *correct* allocation. Nobody verifies the opening hours of a shop nobody visits, and that's fine, because an unverified-but-unconsumed fact causes no harm. The long tail stays uncertified, but "certification proportional to usage value" is a feature, not a bug.

There's a second elegance you didn't mention: **the premium becomes the epistemic signal**. If the underwriting pool will insure "OSM is right about this shop" at 1 cent per dollar, that's a market-priced 99% reliability claim. If the pool demands 8 cents, or

refuses to quote, that itself tells you the entry is suspect — you get a truth-price on every fact *without anyone ever betting on most of them*, just from the pool's quoted underwriting terms. And this structure has real prior art: it's parametric micro-insurance, the model Etherisc used for on-chain flight-delay insurance (automatic payout on an objectively checkable trigger event). Your version is "parametric insurance on database accuracy," which I don't believe anyone has built, and which is a nicer product framing than "prediction market on facts" — including, frankly, for regulators and for ordinary users who understand insurance but not shorting.

Now the honest complications, in rough order of severity:

1. The payout event and the database fact are not the same proposition. The shop can be closed even though OSM is *correct* — one-off illness, public holiday exception, burst pipe. Conversely OSM can be wrong in a way that didn't bite today. If the insurance pays on "shop was closed when I arrived," you're insuring an outcome that includes irreducible real-world risk, and a payout is *not* evidence of a database error — so the dispute/correction machinery shouldn't automatically fire, and the epistemic loop is broken. If instead it pays on "the recorded hours were inaccurate as posted," you now need adjudication of what the shop's actual posted hours were, which is harder to evidence than "door was locked at 3pm." I'd suggest these need to be two explicitly separated products: a parametric convenience product (pays on the experienced outcome, cheap and automatic, no epistemic claim) and a verification bet (pays on the recorded fact being wrong, triggers the correction process). The premium difference between them is itself informative — it prices "the world deviates from the record" separately from "the record is wrong."

2. Evidence quality, in the LLM era, is your hardest engineering problem. A timestamped photo of a closed entrance was decent evidence in 2020. In 2026 it's nearly worthless — generating a photorealistic image of any storefront with any timestamp costs nothing. A \$100 payout is precisely a \$100 bounty on fabricating such evidence, and at scale someone will industrialize it. So the dispute layer needs proof-of-physical-fact machinery: hardware-attested capture (C2PA-signed cameras), location attestations, multiple independent witnesses, or the adjudicator discounting photographic evidence entirely and relying on Schelling-point judgment ("would several independent people checking this shop agree?"). Note this cost only lands on *disputed* claims, so the optimistic structure still protects you — but the fraud economics of the honest-looking-claimant is the attack vector to model first, because unlike oracle capture it scales with thousands of small independent attacks.

3. Moral hazard where the insured can influence the fact. Not much risk with supermarkets, but the general mechanism includes facts individuals control: a shop owner insures against their own listed hours and closes early; someone insures "this Wikidata entry is correct" and then edits it. Insurance has dealt with this for centuries (you can't insure a house you intend to burn — insurable-interest doctrine, exclusions, investigation), but those solutions are identity-heavy and adjudication-heavy, which

fight your pseudonymous, low-cost setting. Minimum viable defense: underwriting excludes claims where the buyer plausibly controls the fact's source, and payout sizes stay small enough that arson-for-profit doesn't pay. This caps the mechanism at micro-insurance scale — which for your use case is probably acceptable.

4. Adverse selection is now the pool's core pricing problem — but that's tolerable.

The person willing to pay \$1-to-win-\$100 disproportionately knows something (saw scaffolding on the shop last week). The pool will systematically lose to informed buyers. This is fine *if* the premium loading covers it — real insurers price adverse selection in every day — but it means the pool needs actuarial data on error rates per fact-type. Opening hours are, notoriously, among the most frequently stale attributes in OSM; a realistic premium there might be 5-10 cents, not 1 cent, and the market teaching everyone that "OSM opening hours are ~92% reliable" would itself be a public service. The pool's loss experience becomes a continuously updated reliability audit of the knowledge base, per attribute type — plausibly the most valuable dataset the whole system generates.

5. Friction and distribution. Almost nobody will consciously buy a \$1 hedge before a grocery run. This only works embedded invisibly at the point of consumption — a maps app or route-planning agent that offers (or auto-attaches) the guarantee when you navigate somewhere. Which suggests the natural first customers aren't humans at all but AI agents: an agent planning a multi-stop errand or a delivery route has exactly the right profile — it consumes thousands of database facts, can price the value of each fact to its plan, transacts at zero psychological cost, and can file evidence-backed disputes automatically. "Machine-to-machine information insurance for agent workflows" might honestly be the wedge product here, with human shoppers coming later.

So: your mechanism works considerably better than my generic bounty-hunter framing, and I'd concede the discovery problem is largely solved *for consumed facts* — which is the right set to solve it for. What it doesn't solve on its own is evidence integrity (problem 2) and the payout-event/fact-proposition split (problem 1); those are where I'd direct design effort, and both belong squarely in the proposal's WP4 risk-analysis matrix. If you pitch this to Krzysztof's team, I'd suggest adding "information

insurance at section — it's the demand: subsidy sink parametric in

Could you explain in more detail the difference between the prediction market and your suggested alternative, the bonded assertion with the optimistic oracle. The problem with existing prediction market seems to be that each bet required capital and most bets will not have a counterparty. But without the counterparty, the bet does not even get made. Maybe what we want is just a credibe offer for the bet, which only allocates capital when there is a counterparty. But is that just what your suggested bonded assertion is? The prediction market bet happens on a smart contract. The bonding of the assertion is also on a smart contract, using the pooled reserve. These sound similar to me but I may be missing something. Is it just an extra level of indirection?

Architected structural distinctions between prediction markets and bonded assertions

Good question, and your intuition — "a credible offer that only allocates capital when a counterparty shows up" — is very close. But the difference is not just indirection; the two mechanisms commit *different amounts* of capital, at *different times*, against *different risks*. Let me take it apart.

What a prediction market bet actually costs

In the standard on-chain design (Gnosis conditional tokens, which Polymarket uses), a position exists only if the full payout is escrowed. Someone deposits \$1, the contract mints one YES and one NO share, and that dollar sits locked until resolution. This is what makes the payout trustless: the winner's \$1 exists in the contract by construction, no matter what anyone does. So a bet has three unavoidable costs:

1. **Full-notional collateral for the entire lifetime.** A \$100 position on a 99% claim locks ~\$100 of combined capital (your \$99, counterparty's \$1) from the moment the bet is struck until resolution — that's the Nagy problem.
2. **A matched counterparty at bet time.** No one on the other side, no position, no price, no information.
3. **Mandatory settlement.** Every market must be resolved by the oracle at expiry, even if nobody disagreed about anything, ever. Each resolution costs a proposer's time, a bond, gas, and a challenge window. For a million mundane facts this cost alone is fatal — you'd run a million adjudications to confirm a million things nobody doubted.

What a bonded assertion costs

In the optimistic pattern, one party posts a claim ("this shop's hours are 9–5") backed by a bond B. Crucially:

1. **Only one side posts capital, and only bond-sized, not notional-sized.** This is the point most easily missed. The bond does not collateralize a payout to a counterparty — it collateralizes the *honesty of the process*. Its size is calibrated to the cost of forcing and running a dispute, not the value riding on the claim. That's why a \$750 UMA bond can secure a market with millions in open interest: the security comes from "anyone in the world can profitably catch a lie during the window," not from escrowed payouts. Bond scales with adjudication cost; market collateral scales with value at stake. Those are wildly different numbers, and the gap between them is exactly your capital efficiency.
2. **The counterparty is latent and universal.** An assertion is a standing, bonded offer to bet, addressed to everyone. You don't need to find a counterparty; the entire world is the counterparty during the challenge window, and their capital is

committed only at the moment of disagreement. Your phrase "a credible offer that only allocates capital when there is a counterparty" describes it almost exactly — with the important note that the offer *is* credible on-chain only because the bond exists. A capital-free offer would be a free option: you could post it, watch the world, and renege. The bond is the minimum stake that makes the standing offer binding. So the assertion is the answer to your own design question: it's the cheapest credible standing bet-offer.

3. **Settlement is lazy.** Undisputed assertions resolve by timeout, for free. The expensive machinery — evidence, adjudication, votes — runs only on the ~1% of claims where someone actually disagrees. Since disagreement is precisely where the information value lives, the system spends adjudication capital exactly where it buys knowledge and nowhere else.
4. **When a dispute does happen, capital escalates in steps, not all at once.** The disputer matches the bond; if the design has multiple rounds (Augur's dispute rounds famously doubled the stake each round), each side commits more only as the disagreement proves itself serious. The "market" materializes on demand, funded to exactly the depth that real disagreement is willing to pay for.

The clean way to see it: two degenerate cases of one mechanism

A prediction market is an assertion system in which every claim is disputed instantly and continuously — both sides fully funded from $t=0$, price discovered at all times. A bonded-assertion system is a prediction market in which the price is administratively pinned at ~1 (or at the asserter's stated value) and trading only occurs if someone pays to cross. They're endpoints of one spectrum, differing along these axes:

	Prediction market	Bonded assertion
Capital committed	Full notional, both sides	One bond, one side
When committed	At bet time	Bond upfront; counterparty capital only on dispute
Sized by	Value at stake	Cost of adjudication
Counterparty	Must be matched	The world, latently
Settlement	Always, per market	Only on dispute (~1%)
Output	Continuous price	Binary status: certified / contested

What you genuinely lose

The assertion side of the table gives up two real things, and this is where your project's design work lives.

You lose the price. An unchallenged assertion tells you "nobody paid to dispute this," which is not a probability. There's no 0.97-vs-0.999 distinction, and no continuous update as evidence shifts. Two ways to recover prices without re-importing full markets: the underwriting premium from our last exchange (the pool's quoted insurance rate on a fact is a price, generated without any bet occurring), or dispute-triggered markets — the assertion sits dormant, and a real two-sided market spins up only when a challenge lands, so price discovery is bought only for contested claims, where it's worth its cost.

You lose odds-expressiveness in the bet itself. Notice a market bet on a 99% claim is naturally asymmetric: \$99 against \$1. A UMA-style dispute is symmetric — bond against equal bond — regardless of the claim's probability. That's a crude contract: challenging a 99.9% claim and challenging a 60% claim cost the same and pay the same. A design opportunity for you: odds-weighted bonds, where the assertion's stated confidence sets the bond ratio (asserting at 99% means your bond faces the challenger's at 99:1). At that point the assertion literally is a limit order — "I offer to bet at these odds, world" — and the two mechanisms have visibly merged. As far as I know, nobody has built an optimistic oracle with asserter-quoted odds; it's a small, concrete, novel mechanism that falls directly out of your framing.

Where the pooled reserve fits — and why it's a second, separate layer

One more distinction worth keeping sharp, because the two capital layers in your system have different jobs and different sizing math. The **bond layer** secures process honesty and scales with adjudication cost — it can be small and shared aggressively across claims, because bonds are only ever at risk on the rare disputed claim. The **reserve layer** (your fractional-reserve pool) backs actual *payouts* — the \$100 the shopper collects — and scales with notional exposure times loss probability times correlation. The escrow model collateralizes payouts at 100%; your pool collateralizes them fractionally against actuarial reserves. So the full architecture is: assertions bonded from a shared bond pool (cheap, process security), payouts underwritten from an actuarially sized reserve (the insurance business), and full capital deployment occurring only in the rare, valuable event of genuine disagreement. That's not an extra level of indir

and adjudica What would the design for a software implementation of your suggestion
and pay for c look like? Let's first just discuss the design.

Architected modular system design spanning claims, disputes, oracles, and evidence layers

Let's design it top-down. I'll describe the system as layered modules, then the state machine of a claim (which is where the real design decisions live), then the parameters and the open choices you'd have to make. I'll keep it implementation-agnostic (EVM assumed, but nothing depends on it).

The layer stack

Five layers, deliberately decoupled so each can be replaced or upgraded independently:

1. Claim layer — the semantic anchor. A registry mapping a `claimId` to a precise, versioned, machine-interpretable proposition. This layer holds no money and is the most underrated part of the whole system: essentially every real-world prediction-market dispute is about wording, not facts, so the claim schema is your primary defense. A claim record needs: the subject reference (a Wikidata QID + property + statement hash, or an OSM element ID + version number — always a *versioned* pointer, never "whatever the database says now"), the proposition semantics (a claim *type* from a small controlled vocabulary, not free text: `attribute-matches-source`, `attribute-matches-world`, `entity-exists`, etc.), the temporal scope (asserted-as-of timestamp, validity window), and the resolution procedure (which evidence types are admissible, which adjudicator escalation path applies). Free-text claims are a v2 luxury; v1 should only admit claims machine-generated from templates over KB triples, because that's what makes millions of claims cheap and disputes crisp. Note the distinction we established earlier must be encoded here as two claim types: "the record matches the world" vs. "the world will match my experience" — same subject, different products, different resolution.

2. Assertion layer — the optimistic core. The engine that attaches bonded truth-status to claims. This is a generalization of UMA's OO v3, with the modifications we discussed. An assertion is `(claimId, asserter, confidence, bondRef, livenessWindow, status)`. Two deliberate departures from stock UMA: assertions carry a **stated confidence**, which sets asymmetric dispute odds (asserting at 99% means a challenger risks 1 to your 99, pro-rated on bond size) and doubles as the system's price signal; and bonds are **references into a shared bond pool** rather than per-assertion escrow — more below. Undisputed assertions resolve to `Certified` by timeout at zero marginal cost. This layer exposes the system's main read API: for any claim, `status ∈ {Unasserted, Asserted, Contested, Certified, Refuted, Expired}` plus the confidence and the size of capital standing behind it.

3. Capital layer — two pools with different math. Keep these strictly separate in the design, because conflating them is the classic error. The **bond pool** is an underwriter-of-process: LPs deposit yield-bearing collateral (sDAI-style, so idle capital earns — this is your fix for the 99-cent problem applied at the pool level), and the pool programmatically backs assertions on claims matching its policy. Its exposure per claim is bond-sized; its risk is "how often are our assertions successfully refuted," which for template-generated claims from a decent KB should be rare and mostly idiosyncratic — *except* for the systemic adjudicator-failure term, which must be modeled as a correlated shock. The **payout reserve** is the insurance business: it sells the consumer product (the \$1-for-\$100 shopper hedge), holds actuarial reserves sized by $\text{notional} \times \text{loss rate} \times \text{correlation loading}$, and its solvency logic is Cramér–

Lundberg-style with explicit per-fact-type loss experience tables that it *learns from its own claims history*. The bond pool can be one contract with a policy whitelist; the reserve wants tranching eventually (junior/senior) but v1 can be a single mutual with a hard exposure cap per claim and per claim-cluster.

4. Adjudication layer — pluggable, escalating. A claim's resolution procedure names an escalation path, e.g.: automated evidence check (an oracle bot verifies a signed data feed or a C2PA-attested photo's provenance) → staked human panel (Kleros-style drawn jurors) → full token-holder vote as the court of last resort. Design principle: the cheap rungs handle volume, the expensive rungs exist mainly as deterrence and are almost never invoked. This layer must be an interface, not an implementation — you will want to swap adjudicators per claim domain (OSM facts vs. Wikidata facts need different evidence norms) and upgrade as evidence tech evolves. The evidence itself (photos, attestations, documents) goes to content-addressed decentralized storage with only hashes on-chain — Swarm or IPFS both fit; given your circle, Swarm's postage-stamp persistence model is arguably a natural fit for evidence that must remain retrievable for the life of a dispute.

5. Graph layer — the netting engine (deferred, but designed-for from day one). The generalized-negRisk component: a constraint store recording logical relations between claims (`A implies B`, `A excludes B`, `exactly-one-of {A1...An}`) in a restricted language you've proven tractable, plus a collateral calculator that computes worst-case exposure over the feasible-outcome polytope instead of summing per-claim. Critically: the interfaces of layers 2–3 should accept a *portfolio* collateral requirement from day one even if v1's calculator is the trivial per-claim sum. That way the graph layer slots in later without migrating capital.

The claim lifecycle (where the design decisions concentrate)

```
Unasserted → Asserted → (liveness expires) → Certified → (validity expires) → Stale
                                     ↓ dispute
                               Contested → adjudication → Certified (challenger slashed)
                                                         → Refuted (asserter slashed, KB
correction event emitted)
```

The decisions hiding in this diagram:

Who may assert, and what stops assertion spam? Anyone, but assertions cost a fee that accrues to the bond pool — and here's the elegant loop: for the bulk case, the *pool itself* is the asserter, auto-asserting over whole KB slices per its policy (e.g. "all OSM opening_hours in Prague, confidence 0.93, bond 0.5 DAI each"). Individual asserters matter at the margins: someone with private knowledge asserting *against* the pool's blanket confidence, or asserting claims the pool declines. The pool's *refusal* to assert a claim is itself signal.

What does a dispute unlock? Minimal version: symmetric-in-odds bond match, single adjudication, done. Richer version (recommended): a dispute *opens a market* — the contested claim spins up a real two-sided order book or small AMM for the duration of the dispute window, so third parties with information can take positions, and the adjudicator's eventual ruling settles them. This is the "market on demand" idea made concrete: price discovery is purchased only for the ~1% of claims where disagreement proves someone will pay for it. Design note: the dispute-market must settle on the *adjudicator's ruling*, which means the adjudicator's incentive-compatibility now carries trading stakes too — size caps on dispute-markets in v1.

What happens on **Refuted?** The contract can't fix Wikidata. It emits a **CorrectionAssertion** event carrying the evidence hash and the adjudicated truth — an oracle *output* that downstream consumers (bots that propose KB edits, apps that overlay corrections) subscribe to. Important honesty point for the design docs: the system produces an authenticated correction *feed*, not corrected databases; the last mile is social.

What does **Certified license?** This is where the consumer product attaches. The insurance module sells hedges only against claims in **Asserted** or **Certified** status, prices them off the assertion confidence plus its own loss tables, and — key wiring — a paid insurance claim on a **attribute-matches-world** product *automatically funds and files* a dispute on the sibling **attribute-matches-source** claim when the evidence suggests the record (not just the day) was wrong. That's the coupling that turns shoppers into your verification workforce without them ever thinking about it.

Parameters you must choose (and my priors)

Bond size: calibrated to adjudication cost at the cheapest escalation rung, not claim value — order of \$1–10 for template claims. Liveness window: long for cold claims (days-weeks; nobody's watching, so give the world time), but note the insurance coupling shortens effective discovery time to "next consumption event," which is the real safety mechanism. Confidence granularity: coarse buckets (0.9 / 0.97 / 0.99 / 0.999), because false precision invites gaming of the odds ratio. Slashing split: majority of the loser's stake to the winner, a slice burned or to the adjudication treasury (anti-collusion: prevents self-dispute laundering being free). Per-cluster exposure caps in the reserve: derived from the constraint graph — claims sharing a source, a region, or an adjudicator are one cluster.

What I would explicitly *not* build in v1

No continuous markets on uncontested claims. No leverage/liquidation machinery anywhere. No free-text claims. No cross-chain anything. No token — fees and yield are sufficient incentive plumbing to test the mechanism, and adding a token before the loss tables exist just adds a reflexive attack surface.

The build order that de-risks fastest

Phase 0 is a simulation, not contracts: agent-based model (cadCAD or plain Python) of the assertion/dispute/insurance loop over a synthetic KB with planted errors — the deliverable is loss tables and the answer to the go/no-go question, "at what dispute cost and consumption rate does planted-error half-life become acceptable?" Phase 1 is layers 1-2 plus a naive bond pool on a testnet against a frozen snapshot of one narrow domain (I'd pick OSM `opening_hours` in one city: high error base rate, cheap physical verification, crisp evidence). Phase 2 adds the insurance product and the dispute-triggered mini-market. Phase 3 is the graph layer, once you have real correlation data to make netting mean something.

The two hardest open design problems, named plainly so they land in the right work package: the **evidence-admissibility spec** for the automated adjudication rung (what counts as proof a shop was closed, in a world of generative imagery — this is a moving target and should be a versioned, per-domain policy document, not code), and the **odds-weighted bond game theory** (asymmetric bonds create a new strategic surface — e.g., asserting at 0.999 to make challenges maximally expensive — which needs an equilibrium analysis before you trust it with real stakes; nobody has published this, so it's simultaneously your risk and your paper).

That's the shape. Want to zoom into any one layer next — I'd suggest either the claim schema (most consequential, least glamorous) or the two-pool capital accounting (where the solvency invariants live)?

